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INSTITUTION	JECRC College
BRANCH / BATCH	B.Tech CSE · 2027 batch
PROBLEM STATEMENT	Digital twin of transformer or any other electrical equipment

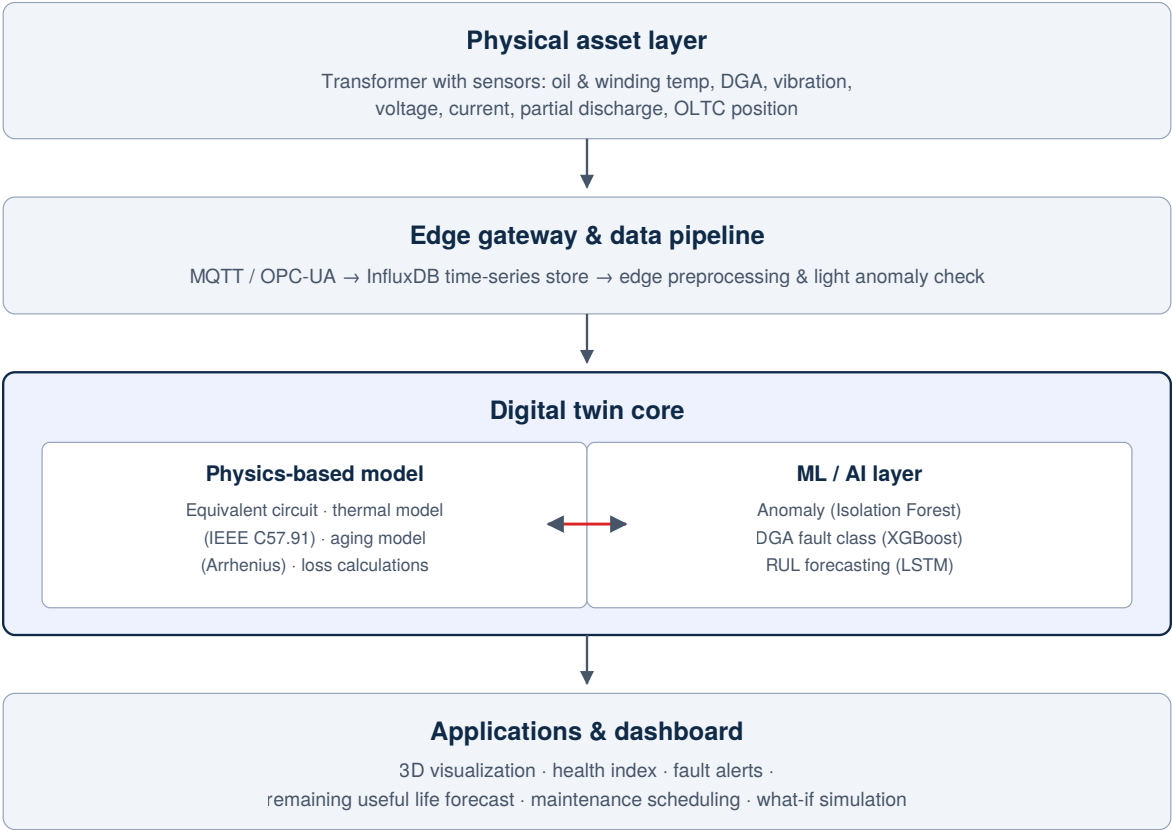
Inputs Considered

Four input categories the digital twin needs to faithfully mirror, reason about, and forecast equipment behavior.

01 Real-time sensor streams Oil temperature (top & bottom), winding hotspot temperature, primary/secondary voltage and current, Dissolved Gas Analysis (H₂, CH₄, C₂H₂, C₂H₄, C₂H₆, CO, CO₂), vibration spectra, partial discharge counts, OLTC tap position, ambient temperature and humidity. Why → Live vital signs that drive every prediction the twin makes about the asset's present state.	02 Static asset data Nameplate ratings (kVA, voltage class, frequency), per-unit impedance, cooling class (ONAN / ONAF / OFAF), insulation class, winding configuration, OEM design specifications, geometric and material parameters from the transformer datasheet. Why → Calibrates the physics model to this specific asset rather than a generic transformer.
03 Historical operational data Load profiles across seasons, switching events, overload incidents, past maintenance records, fault history, repair logs, and any prior alarm events. Time-aligned with weather and grid-loading data where available. Why → Training signal for ML models and baseline distribution for anomaly detection.	04 Domain priors & standards IEEE C57.91 loading guide for hotspot & aging, IEC 60599 for DGA interpretation, the Duval Triangle and Rogers Ratio methods for fault typing, and the Arrhenius equation for paper-insulation thermal aging. Why → Encodes 70+ years of transformer engineering. Beats purely data-driven models when training data is limited.

Process — Architecture

A four-tier system: physical asset → edge ingestion → digital twin core (physics + ML) → operator-facing applications.



Data flow

Upward: IoT sensors stream measurements through an edge gateway using industrial protocols (MQTT, OPC-UA). Edge nodes do lightweight preprocessing and a first-pass anomaly check before forwarding to the cloud time-series store.

Twin reasoning: The digital twin core fuses two complementary engines. The physics model predicts *expected* behavior under current loading and ambient conditions; the ML layer interprets *deviations* from expectation and forecasts future degradation. Neither approach alone is sufficient — together they outperform either.

Downward: Insights flow back through dashboards and APIs into operator workflows: real-time alerts, condition-based maintenance schedules, and "what-if" load simulations. The architecture integrates natively with **ABB Ability™** on Microsoft Azure Digital Twins.

TECH STACK

MQTT

OPC-UA

InfluxDB

Python · FastAPI

PyTorch

scikit-learn

XGBoost

Three.js

Plotly

Azure Digital Twins

Docker · Kubernetes

Process — Physics-Based Modeling

Three physics-grounded sub-models compute what the transformer should be doing at every instant. ML detects when reality deviates.

1. Thermal model

IEEE C57.91

Predicts the winding hotspot temperature from load, ambient temperature, and cooling state using the IEEE C57.91 loading guide. Output is compared to the measured hotspot reading: persistent deviation indicates cooling-system degradation, blocked oil flow, or incipient insulation failure long before hard thresholds are breached.

$$\theta_h = \theta_a + \Delta\theta_{tor} + \Delta\theta_{hs}$$

Inputs: load current, ambient temp, cooling class → **Output:** predicted hotspot θ_h (°C)

2. Aging & remaining-life model

ARRHENIUS

Insulation paper degrades exponentially with temperature. The Arrhenius equation translates the predicted hotspot trajectory into a "loss-of-life per hour" curve. Integrated over operating history, this gives total insulation life consumed and a physics-grounded prior for the ML-based RUL forecast on the next page.

$$\text{Rate} = A \cdot \exp(-E_a / R \cdot T)$$

Inputs: hotspot history, activation energy → **Output:** life consumed (%), aging acceleration factor

3. Electrical equivalent circuit

EQUIVALENT CKT

A standard T-model with primary/secondary winding resistance, leakage reactance, and magnetizing branch. Used for two purposes: computing real-time losses (copper + iron) and simulating "what-if" fault scenarios — inter-turn short, ground fault, magnetizing inrush — without touching the physical asset. Modeled in Python with SciPy ODE solvers; can be ported to MATLAB/Simulink or Modelica.

Inputs: primary V/I, secondary V/I, impedance parameters → **Output:** losses, fault simulation traces, efficiency map

Validation strategy

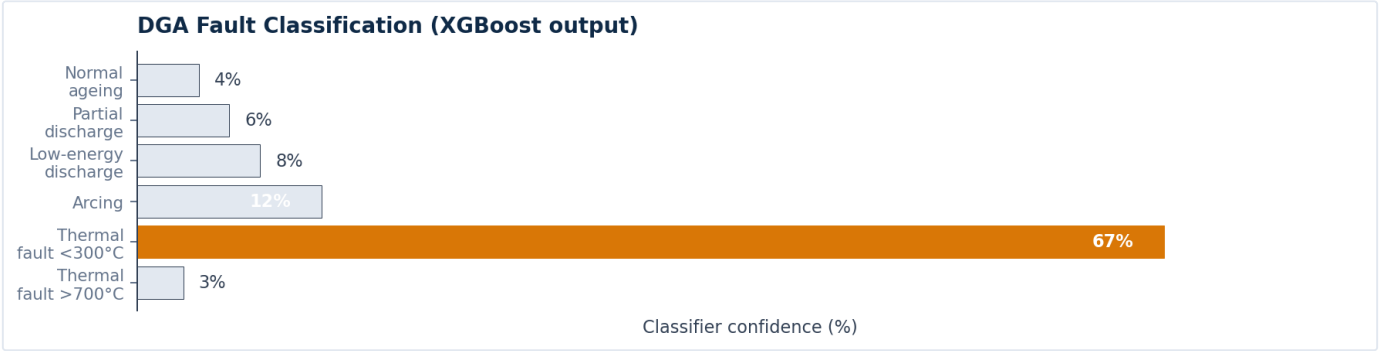
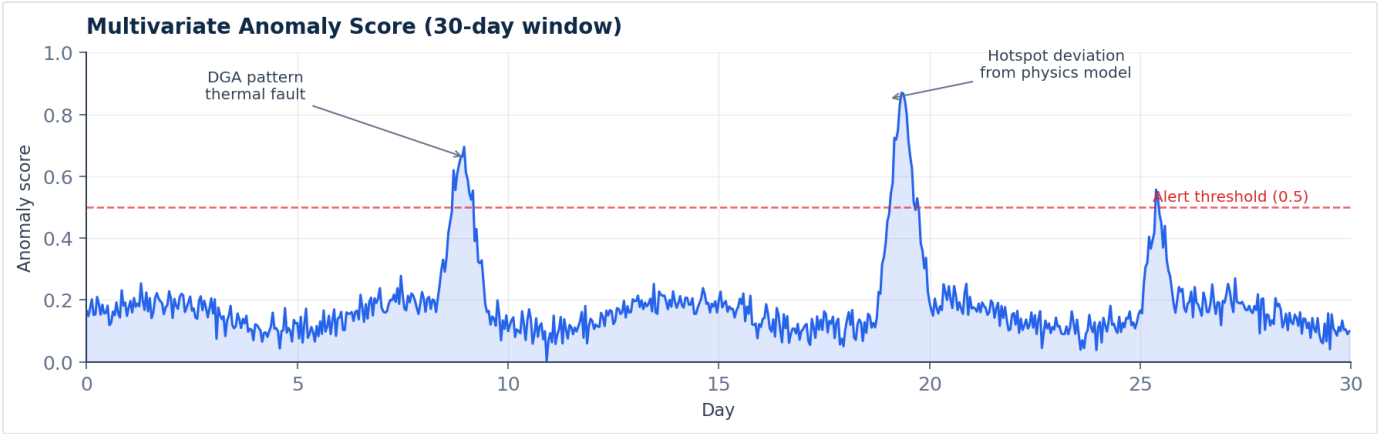
Models are calibrated against the OEM datasheet and refined using historical sensor data via least-squares fitting. Where the physics model and measurements disagree consistently, the parameters are re-estimated (online Kalman filter approach for thermal time constants). This keeps the "expected behavior" baseline accurate as the asset ages and operating regimes shift over time.

Process — ML/AI Layer

Three complementary ML models interpret what the physics model can't: pattern-rich anomalies, fault categories, and time-to-maintenance.

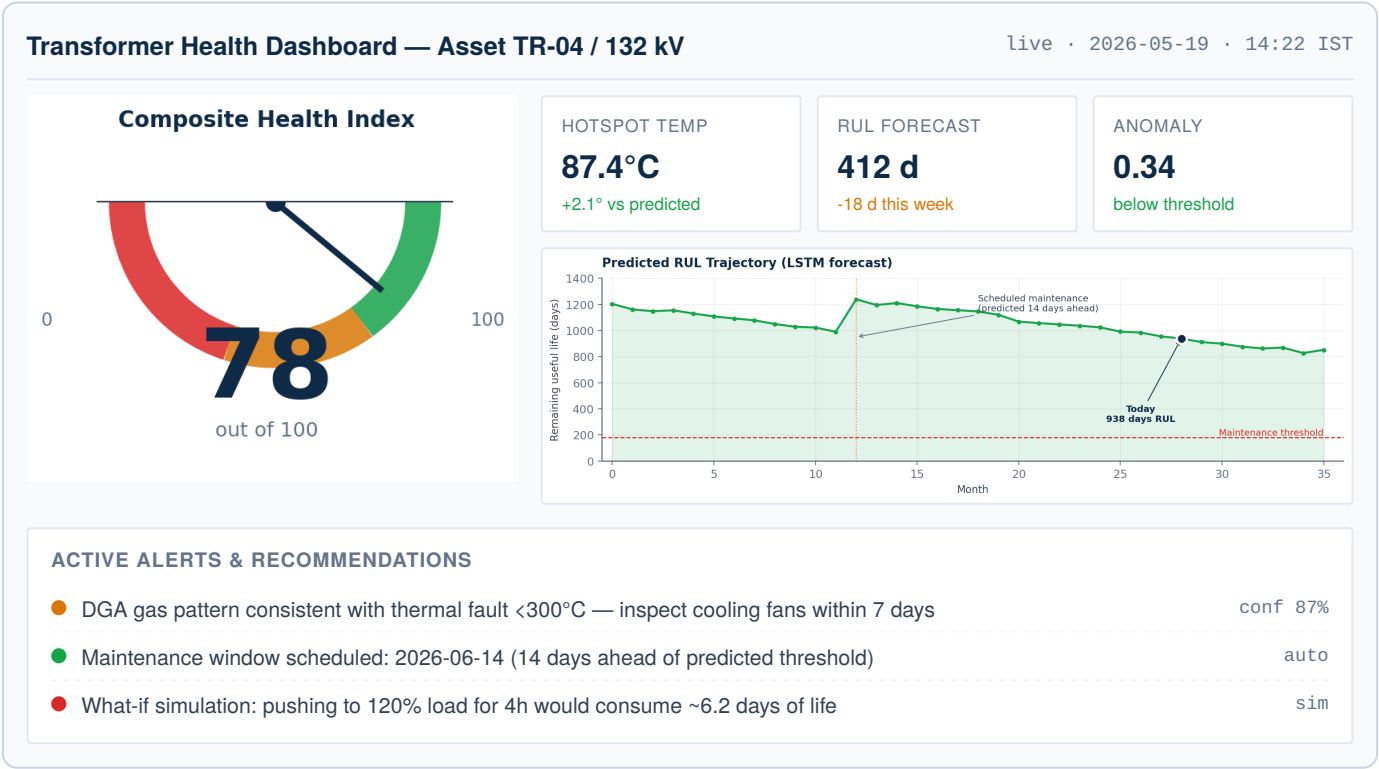
Anomaly detection	Fault classification	RUL forecasting
INPUT Multivariate sensor vector + physics residuals (24-dim)	INPUT DGA gas ratios + Duval Triangle coordinates	INPUT Time-series of degradation indicators (90-day window)
MODEL Isolation Forest + LSTM autoencoder (ensemble)	MODEL XGBoost classifier (6-class)	MODEL LSTM with attention / Temporal Fusion Transformer
OUTPUT Anomaly score $\in [0,1]$ every 60s	OUTPUT Fault type + confidence + SHAP explanations	OUTPUT Remaining useful life in days + prediction interval
USE First-line alert — "this state is unusual given current loading"	USE Diagnose root cause: thermal, PD, arcing, or normal aging	USE Drives condition-based maintenance scheduling

THE HYBRID LOOP The physics model predicts expected behavior under current loading. The ML layer interprets the *residual* — the gap between expected and observed — to flag faults and forecast life. Neither approach alone matches this fidelity: pure physics misses unmodeled degradation modes, while pure ML demands prohibitive training data. The combination is how industrial-grade digital twins actually work.



Expected Output & Business Value

A live operator dashboard plus quantified business outcomes. Architecture extends beyond transformers.



ACTIVE ALERTS & RECOMMENDATIONS

DGA gas pattern consistent with thermal fault <300°C — inspect cooling fans within 7 days

conf 87%

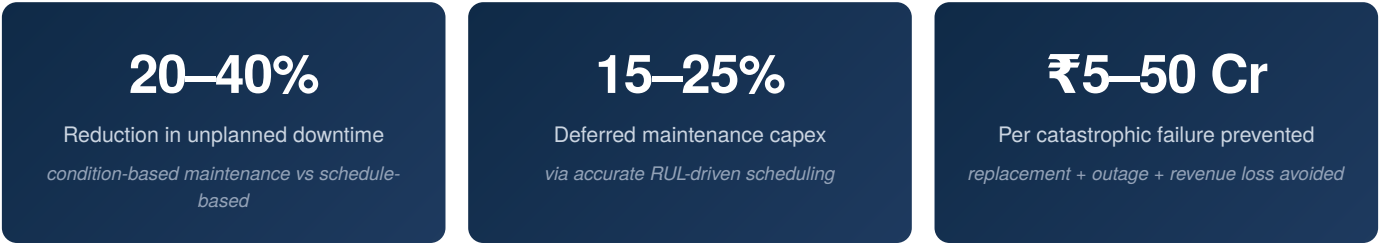
Maintenance window scheduled: 2026-06-14 (14 days ahead of predicted threshold)

auto

What-if simulation: pushing to 120% load for 4h would consume ~6.2 days of life

sim

Business value



Scalability → The architecture generalizes from transformers to **generators, exciters, and large motors**: swap the physics module to the corresponding equipment model — the edge pipeline, data layer, ML layer, and operator dashboard are asset-agnostic. A single deployment template scales horizontally across an ABB customer's entire substation or plant fleet.

Submitted for ABB EngineeredX 2.0 · Problem Statement 1 · May 2026